



UNIVERSITÉ
**PARIS
SCIENCES
& LETTRES**

Statistical Learning II

Lecture 3 - Supervised learning (continued)

Bruno Loureiro
@ CSD, DI-ENS & CNRS

bruno.loureiro@di.ens.fr

Empirical risk

Let $\mathcal{D} = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote the **training data**.

Given a loss function $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}_+$, and a predictor $f : \mathcal{X} \rightarrow \mathcal{Y}$ define the **empirical risk**:

$$\hat{\mathcal{R}}_n(f) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i))$$

Also known as the **training loss**. This quantifies how well we fit the data. But is this a good notion of learning?

$$f(x) = \begin{cases} y_i & \text{if } x \in \mathcal{D} \\ 0 & \text{otherwise} \end{cases} \quad \Rightarrow \quad \hat{\mathcal{R}}_n = 0$$

Probabilistic framework

Instead, we would like predictors that do well on **unseen data**.

Probabilistic framework

Instead, we would like predictors that do well on **unseen data**.

Assume there is an underlying data distribution p over $\mathcal{X} \times \mathcal{Y}$:

$$(x_i, y_i) \sim p \quad \text{i.i.d.}$$

Probabilistic framework

Instead, we would like predictors that do well on **unseen data**.

Assume there is an underlying data distribution p over $\mathcal{X} \times \mathcal{Y}$:

$$(x_i, y_i) \sim p \quad \text{i.i.d.}$$



- The “i.i.d.” assumption might not always hold. (Sampling bias, distribution shift, etc.)
- Under this assumption, $\hat{\mathcal{R}}_n$ is a random function.

Population risk

Instead, we would like predictors that do well on **unseen data**.

Assume there is an underlying data distribution p over $\mathcal{X} \times \mathcal{Y}$:

$$(x_i, y_i) \sim p \quad \text{i.i.d.}$$

Define the notion of population risk of a predictor $f: \mathcal{X} \rightarrow \mathcal{Y}$:

$$\mathcal{R}(f) = \mathbb{E} [\ell(y, f(x))]$$

Also known as the **generalisation** or **test error**.

Population risk

Instead, we would like predictors that do well on **unseen data**.

Assume there is an underlying data distribution p over $\mathcal{X} \times \mathcal{Y}$:

$$(x_i, y_i) \sim p \quad \text{i.i.d.}$$

Define the notion of population risk of a predictor $f: \mathcal{X} \rightarrow \mathcal{Y}$:

$$\mathcal{R}(f) = \mathbb{E} [\ell(y, f(x))]$$

Also known as the **generalisation** or **test error**.



$\mathcal{R}$ is a deterministic function of the predictor f

Validation set

In practice, the statistician almost never has access to the data distribution.

A common procedure to estimate $\mathcal{R}$ consists of splitting the training data in **training** and **validation** set $\mathcal{D} = \mathcal{D}_T \cup \mathcal{D}_V$.

Train on $\mathcal{D}_T$, test on $\mathcal{D}_V$.

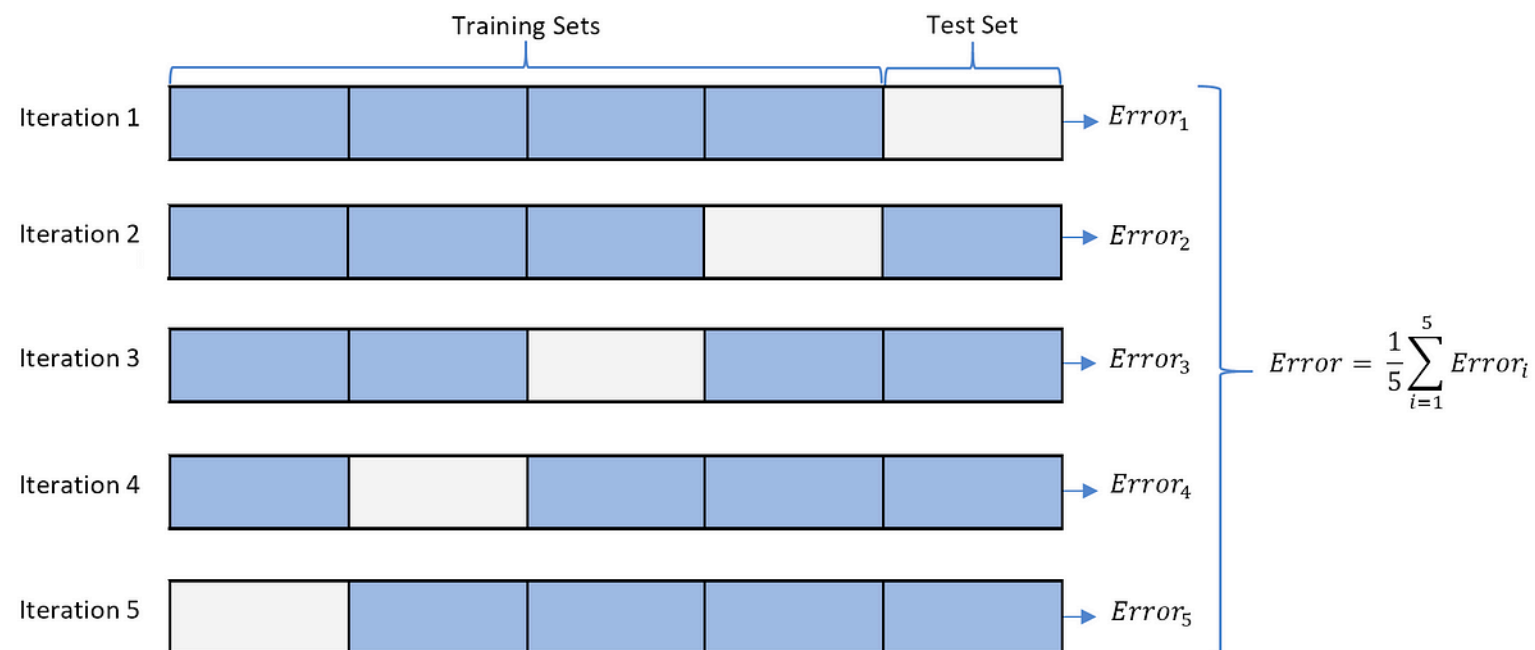
Validation set

In practice, the statistician almost never has access to the data distribution.

A common procedure to estimate $\mathcal{R}$ consists of splitting the training data in **training** and **validation** set $\mathcal{D} = \mathcal{D}_T \cup \mathcal{D}_V$.

Train on $\mathcal{D}_T$, test on $\mathcal{D}_V$.

To reduce error, often one repeats this procedure k times, averaging over the result. This is known as **k fold cross-validation**.



Conditional risk

Given the data distribution p and a loss function ℓ , we can decompose the risk:

$$\mathcal{R}(f) = \mathbb{E}_{(X,Y) \sim p}[\ell(Y, f(X))]$$

Conditional risk

Given the data distribution p and a loss function ℓ , we can decompose the risk:

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(X,Y) \sim p}[\ell(Y, f(X))] \\ &= \mathbb{E}_{X \sim p_x} \left[\mathbb{E}[\ell(Y, f(x)) \mid X = x] \right]\end{aligned}$$



The internal expectation is over the conditional distribution $Y|X = x$

Conditional risk

Given the data distribution p and a loss function ℓ , we can decompose the risk:

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(X,Y) \sim p}[\ell(Y, f(X))] \\ &= \mathbb{E}_{X \sim p_x} \left[\mathbb{E}[\ell(Y, f(x)) \mid X = x] \right]\end{aligned}$$

“Conditional risk”

Conditional risk

Given the data distribution p and a loss function ℓ , we can decompose the risk:

$$\begin{aligned}\mathcal{R}(f) &= \mathbb{E}_{(X,Y) \sim p}[\ell(Y, f(X))] \\ &= \mathbb{E}_{\textcolor{red}{X} \sim p_x} \left[r(z \mid \textcolor{red}{X}) \right] \quad z = f(x) \in \mathcal{Y}\end{aligned}$$

Where we have defined:

$$r(z \mid x) = \mathbb{E}[\ell(Y, z) \mid X = x]$$

“Conditional risk”

Bayes risk

The Bayes predictor is the best achievable predictor:

$$f_{\star}(x) \in \operatorname{argmin}_{z \in \mathcal{Y}} r(z | x)$$

And is also known as the **target function**.

Bayes risk

The Bayes predictor is the best achievable predictor:

$$f_{\star}(x) \in \operatorname{argmin}_{z \in \mathcal{Y}} r(z | x)$$

And is also known as the **target function**.

Similarly, the **Bayes risk** is the best achievable risk:

$$\mathcal{R}_{\star} = \mathbb{E}_{X \sim p_x} \left[\inf_{z \in \mathcal{Y}} r(z | X) \right]$$

Bayes risk

The Bayes predictor is the best achievable predictor:

$$f_{\star}(x) \in \operatorname{argmin}_{z \in \mathcal{Y}} r(z | x)$$

And is also known as the **target function**.

Similarly, the **Bayes risk** is the best achievable risk:

$$\mathcal{R}_{\star} = \mathbb{E}_{X \sim p_x} \left[\inf_{z \in \mathcal{Y}} r(z | X) \right]$$



- The Bayes predictor $f_{\star}$ might not be unique.
- Typically we have $\mathcal{R}_{\star} \neq 0$. 🤔 Examples in the TD

Learning algorithm

Let $\mathcal{D}_p = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

A **learning algorithm** is a map that takes the training data and returns a predictor

$$\mathcal{A} : \mathcal{D}_p \mapsto f$$

Learning algorithm

Let $\mathcal{D}_p = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

A **learning algorithm** is a map that takes the training data and returns a predictor

$$\mathcal{A} : \mathcal{D}_p \mapsto f$$

You have seen many examples in “*Statistical Learning I*”:

- K-nearest neighbours
- Decision trees
- Random Forests
- Least-squares regression

No free lunch

Consider a binary classification task with $\mathcal{Y} = \{0,1\}$ and 0/1 loss $\ell(y, z) = \delta_{yz}$. Let $\mathcal{P}$ denote the set of probability distributions over $\mathcal{X} \times \{0,1\}$.

Theorem

For any $n \in \mathbb{N}$ and algorithm $\mathcal{A}$ over $(\mathcal{X} \times \{0,1\})^{\otimes n}$, there exists $p \in \mathcal{P}$ such that

$$\sup_{p \in \mathcal{P}} \left\{ \mathbb{E} \left[\mathcal{R}(\mathcal{A}(\mathcal{D}_p)) \right] - \mathcal{R}_\star \right\} \geq \frac{1}{2}$$

No free lunch

Consider a binary classification task with $\mathcal{Y} = \{0,1\}$ and 0/1 loss $\ell(y, z) = \delta_{yz}$. Let $\mathcal{P}$ denote the set of probability distributions over $\mathcal{X} \times \{0,1\}$.

Theorem

For any $n \in \mathbb{N}$ and algorithm $\mathcal{A}$ over $(\mathcal{X} \times \{0,1\})^{\otimes n}$, there exists $p \in \mathcal{P}$ such that

$$\sup_{p \in \mathcal{P}} \left\{ \mathbb{E} \left[\mathcal{R}(\mathcal{A}(\mathcal{D}_p)) \right] - \mathcal{R}_\star \right\} \geq \frac{1}{2}$$

In words: For any algorithm you choose, one can always construct a data distribution such that your error is at best equal than random guessing.

Empirical risk minimisation

Let $\mathcal{D} = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

Empirical risk minimisation (ERM) is a class of learning algorithms that consist of minimising the empirical risk:

$$\min_f \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) \quad (= \hat{\mathcal{R}}_n(f))$$

Empirical risk minimisation

Let $\mathcal{D} = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

Empirical risk minimisation (ERM) is a class of learning algorithms that consist of minimising the empirical risk:

$$\min_f \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) \quad (= \hat{\mathcal{R}}_n(f))$$



By the law of large numbers, for a given f

$$\hat{\mathcal{R}}_n(f) \xrightarrow{P} \mathcal{R}(f) \quad \text{as} \quad n \rightarrow \infty$$

However, at fixed n , $\hat{\mathcal{R}}_n$ can be very different from $\mathcal{R}$

Empirical risk minimisation

$$\min_f \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) \quad (= \hat{\mathcal{R}}_n(f))$$

ERM maps supervised learning to an optimisation problem.

Empirical risk minimisation

$$\min_f \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) \quad (= \hat{\mathcal{R}}_n(f))$$

ERM maps supervised learning to an **optimisation problem**.

But optimising on the space of functions is computationally intractable....

Empirical risk minimisation

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f(x_i)) \quad (= \hat{\mathcal{R}}_n(f))$$

ERM maps supervised learning to an **optimisation problem**.

But optimising on the space of functions is computationally intractable....

Therefore, we restrict to classes of mathematically and computationally amenable functions:

$$f \in \mathcal{H}$$

Also known as the **hypothesis class**.

Hypothesis class

Most of the time, we consider **parametric classes**.

$$\mathcal{H} = \{f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta \subset \mathbb{R}^p\}$$

Hypothesis class

Most of the time, we consider **parametric classes**.

$$\mathcal{H} = \{f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta \subset \mathbb{R}^p\}$$

Examples: • Linear functions: $f_{\theta}(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{x} \rangle + b$

Hypothesis class

Most of the time, we consider **parametric classes**.

$$\mathcal{H} = \{f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta \subset \mathbb{R}^p\}$$

- Examples:
- Linear functions: $f_{\theta}(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{x} \rangle + b$
 - Generalised Linear functions: $f_{\theta}(\mathbf{x}) = \sigma(\langle \boldsymbol{\theta}, \mathbf{x} \rangle + b)$

Hypothesis class

Most of the time, we consider **parametric classes**.

$$\mathcal{H} = \{f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta \subset \mathbb{R}^p\}$$

- Examples:
- Linear functions: $f_{\theta}(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{x} \rangle + b$
 - Generalised Linear functions: $f_{\theta}(\mathbf{x}) = \sigma(\langle \boldsymbol{\theta}, \mathbf{x} \rangle + b)$
 - Two layer neural network: $f_{\theta}(\mathbf{x}) = \sum_{j=1}^p a_j \sigma(\langle \mathbf{w}_j, \mathbf{x} \rangle + b)$

Hypothesis class

Most of the time, we consider **parametric classes**.

$$\mathcal{H} = \{f_{\theta} : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta \subset \mathbb{R}^p\}$$

- Examples:
- Linear functions: $f_{\theta}(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{x} \rangle + b$
 - Generalised Linear functions: $f_{\theta}(\mathbf{x}) = \sigma(\langle \boldsymbol{\theta}, \mathbf{x} \rangle + b)$
 - Two layer neural network: $f_{\theta}(\mathbf{x}) = \sum_{j=1}^p a_j \sigma(\langle \mathbf{w}_j, \mathbf{x} \rangle + b)$



The choice of hypothesis (or architecture) induces an **inductive bias** in the learning.

e.g. linear functions can only learn linear relationships

Risk decomposition

For any $\theta \in \Theta$, we can decompose the excess risk:

$$\mathcal{R}(\theta) - \mathcal{R}_\star = \left(\mathcal{R}(\theta) - \inf_{\theta' \in \Theta} \mathcal{R}(\theta') \right) + \left(\inf_{\theta' \in \Theta} \mathcal{R}(\theta') - \mathcal{R}_\star \right)$$

Estimation error

Approximation error

Risk decomposition

For any $\theta \in \Theta$, we can decompose the excess risk:

$$\mathcal{R}(\theta) - \mathcal{R}_\star = \left(\mathcal{R}(\theta) - \inf_{\theta' \in \Theta} \mathcal{R}(\theta') \right) + \left(\inf_{\theta' \in \Theta} \mathcal{R}(\theta') - \mathcal{R}_\star \right)$$

- Approximation error $\inf_{\theta' \in \Theta} \mathcal{R}(\theta') - \mathcal{R}_\star$
 - Independent of n
 - Deterministic
 - Typically decreasing with $|\Theta|$ (to 0 if $\mathcal{H}$ rich enough)

Risk decomposition

For any $\theta \in \Theta$, we can decompose the excess risk:

$$\mathcal{R}(\theta) - \mathcal{R}_\star = \left(\mathcal{R}(\theta) - \inf_{\theta' \in \Theta} \mathcal{R}(\theta') \right) + \left(\inf_{\theta' \in \Theta} \mathcal{R}(\theta') - \mathcal{R}_\star \right)$$

- Approximation error $\inf_{\theta' \in \Theta} \mathcal{R}(\theta') - \mathcal{R}_\star$
 - Independent of n
 - Deterministic
 - Typically decreasing with $|\Theta|$ (to 0 if $\mathcal{H}$ rich enough)
- Estimation error $\mathcal{R}(\theta) - \inf_{\theta' \in \Theta} \mathcal{R}(\theta')$
 - Typically Random
 - Typically decreasing with n

Risk decomposition

For any $\theta \in \Theta$, we can decompose the excess risk:

$$\mathcal{R}(\theta) - \mathcal{R}_\star = \left(\mathcal{R}(\theta) - \inf_{\theta' \in \Theta} \mathcal{R}(\theta') \right) + \left(\inf_{\theta' \in \Theta} \mathcal{R}(\theta') - \mathcal{R}_\star \right)$$

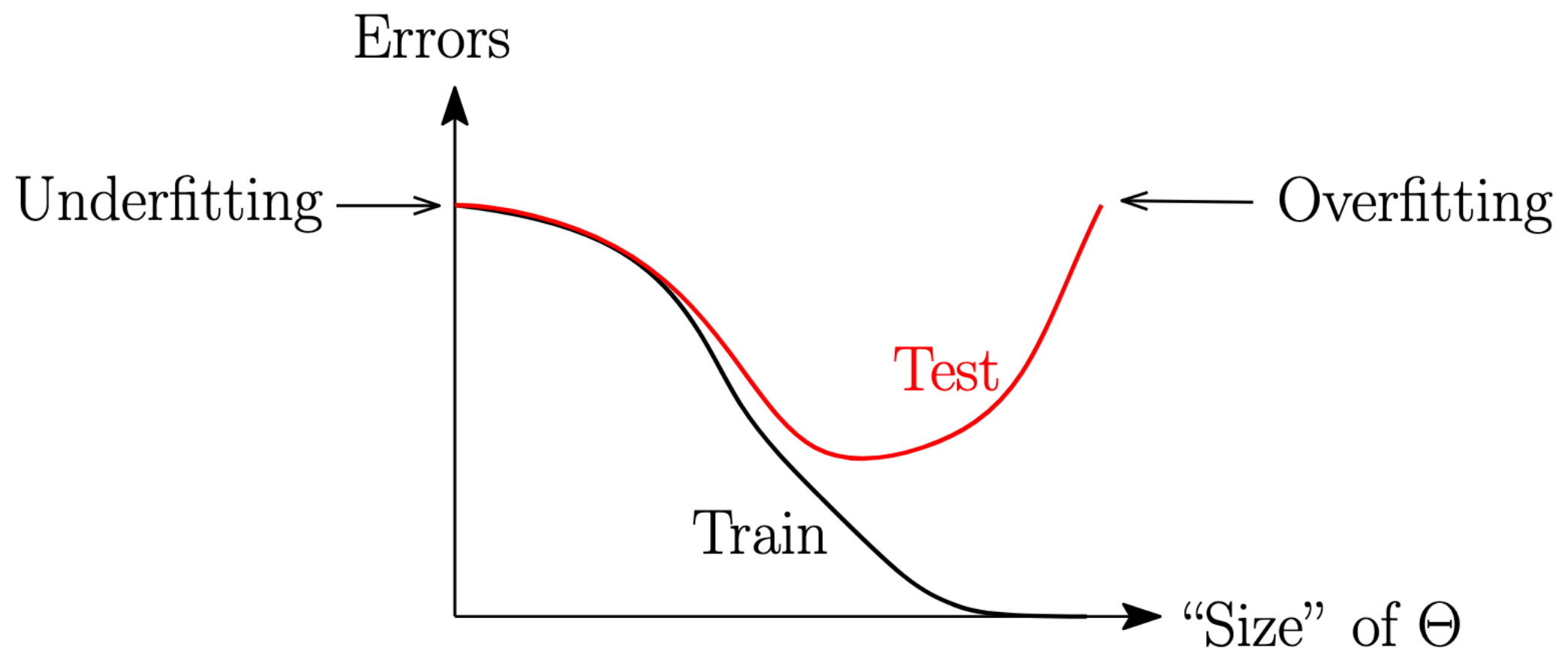


Figure from “Learning Theory from First Principles”, F. Bach 2024

Summary of ERM

Let $\mathcal{D} = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

Given a choice of:

- Parametric hypothesis class $\mathcal{H} = \{f_\theta : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta\}$
- Loss function $\ell : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_+$

Empirical Risk Minimisation consists of:

$$\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_\theta(x_i))$$

Summary of ERM

Let $\mathcal{D} = \{(x_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, n\}$ denote training data sampled i.i.d. from p .

Given a choice of:

- Parametric hypothesis class $\mathcal{H} = \{f_\theta : \mathcal{X} \rightarrow \mathcal{Y} : \theta \in \Theta\}$
- Loss function $\ell : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}_+$

Empirical Risk **Minimisation** consists of:

$$\min_{\theta \in \Theta} \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_\theta(x_i))$$

Key questions

- What optimisation procedure to choose?

$$F(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_{\theta}(x_i))$$

Is typically a non-convex function of $\theta \in \Theta$.

Key questions

- What optimisation procedure to choose?

$$F(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_{\theta}(x_i))$$

Is typically a non-convex function of $\theta \in \Theta$.

- How large n needs to be (with respect to p, d) so that $\hat{\theta} \in \operatorname{argmin} F(\theta)$ has low training and/or test error?

Key questions

- What optimisation procedure to choose?

$$F(\theta) = \frac{1}{n} \sum_{i=1}^n \ell(y_i, f_{\theta}(x_i))$$

Is typically a non-convex function of $\theta \in \Theta$.

- How large n needs to be (with respect to p, d) so that $\hat{\theta} \in \operatorname{argmin} F(\theta)$ has low training and/or test error?
- What properties of the data distribution p makes the problem easier / harder?